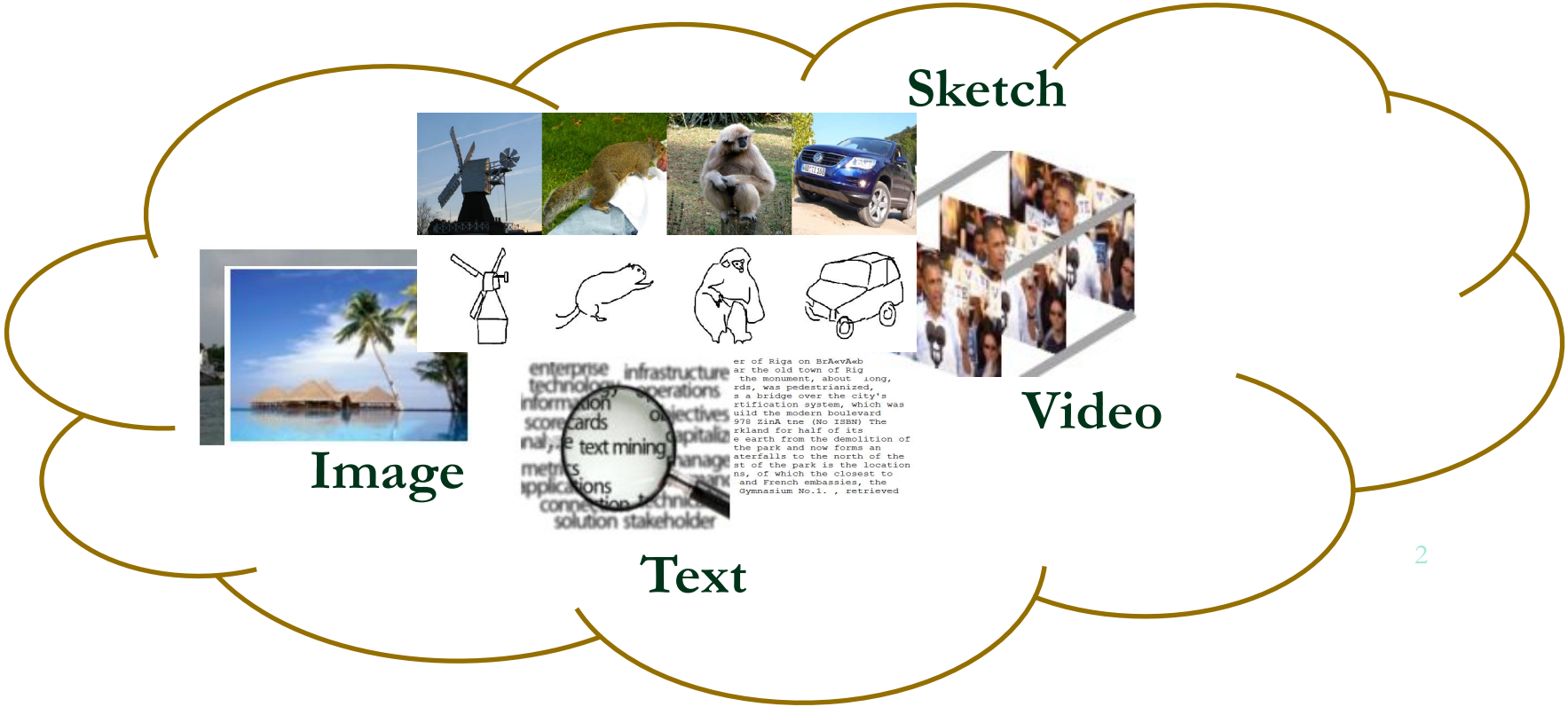
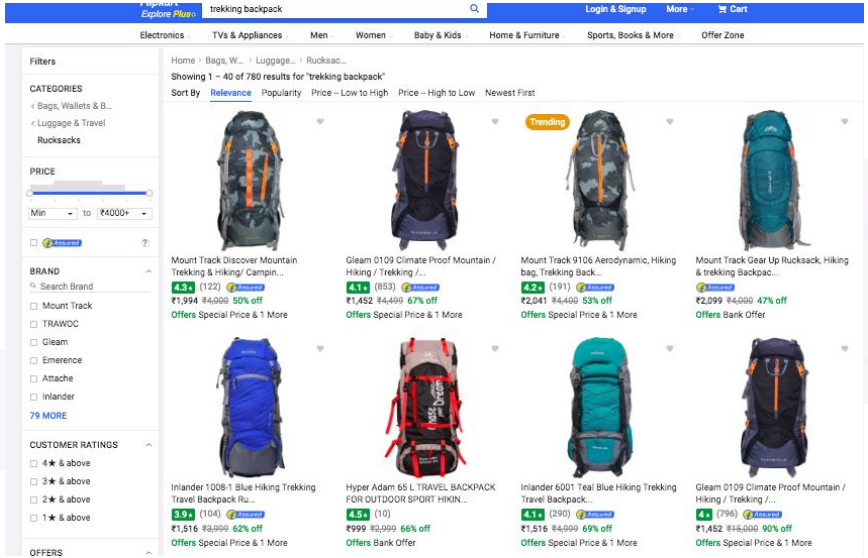
Learning Transferable Visual Models From Natural Language Supervision - CLIP Radford et al.

Soma Biswas

Data stored in different modalities !



Cross-Modal Retrieval



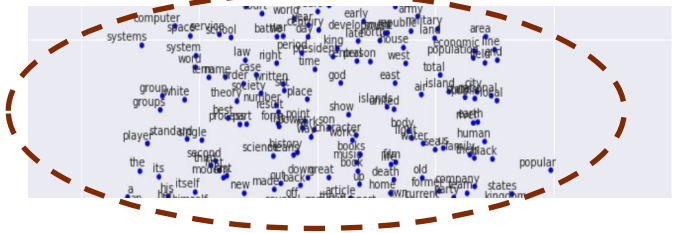
Challenges

1. Significant differences between the different modality
2. Significant variations within the same modality

The monument is located in the center of Riga on BrÃ«vÃ«bas bulvÃ ris ("Freedom Boulevard"), near the old town of Riga. In 1990 a section of the street around the monument, about long, between Raiņa and Aspazija boulevards, was pedestrianized, forming a plaza. Part of it includes a bridge over the city's canal, once a part of the city's fortification system, which was demolished in the 19th century to build the modern boulevard district. RÃ«ga 1860Ã€"1917 RÃ«ga 1978 ZinÃ tne (No ISBN) The canal is long and surrounded by parkland for half of its length., retrieved on 2007-05-30 The earth from the demolition of the fortifications was gathered in the park and now forms an artificial hill with a cascade of waterfalls to the north of the monument. The Boulevard district east of the park is the location of several embassies and institutions, of which the closest to the Freedom Monument are the German and French embassies, the University of Latvia and Riga State Gymnasium No.1. , retrieved on 2007-06-03

Text Query

Search Set images



Text feature-space

How to measure inter-modal relevance of data ?

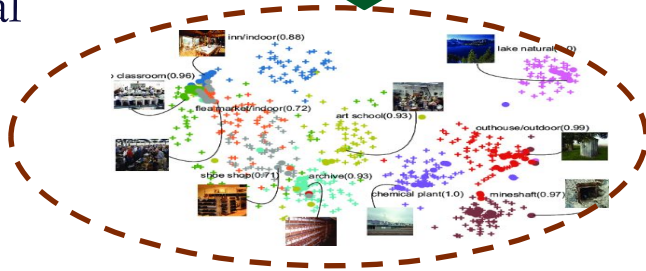
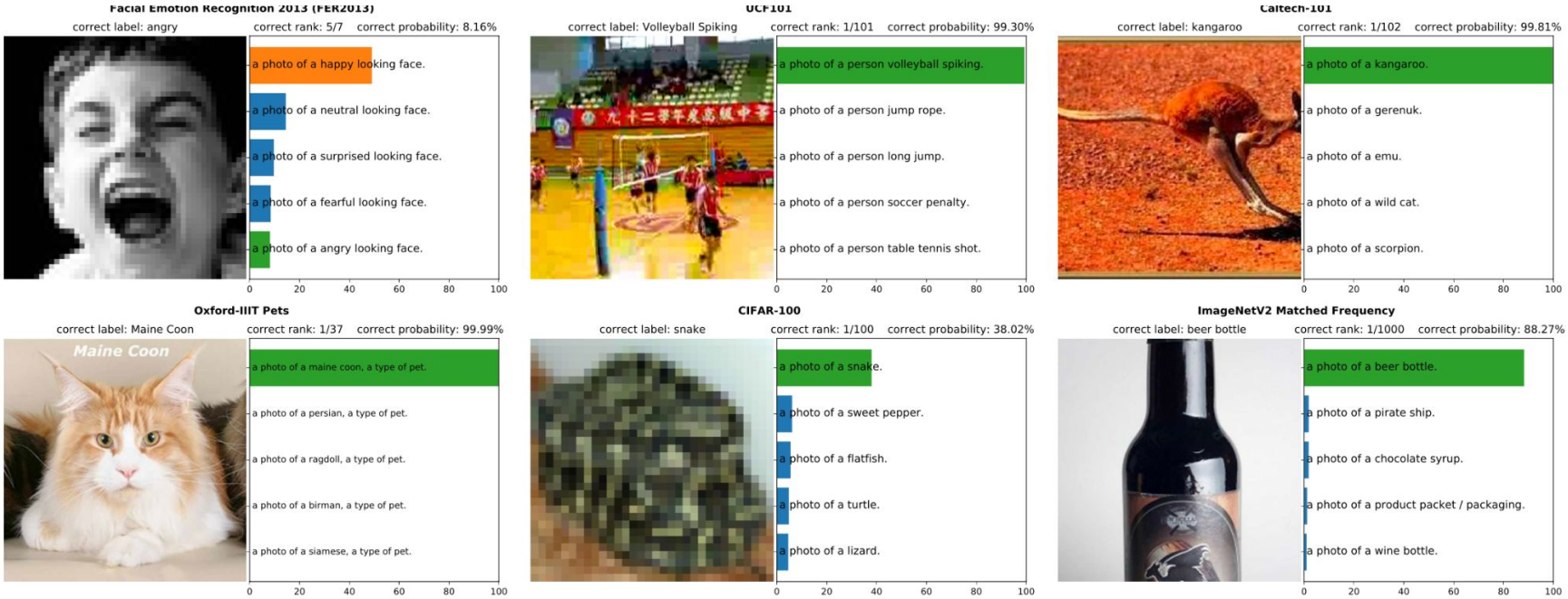


Image feature-space

Map data to a common Latent Space

Few Examples



Main Points

- CLIP is an image classifier
- Contrastive training over predicting which caption goes with which image
- CLIP is a zero-shot classifier
- Beats fully supervised baseline (sometimes by a large margin) on many datasets
- Leverages a huge amount of paired data
- Approach isn't *entirely* novel, but never been done at this scale/efficiency or with the huge number of experiments

- Existing image classification models...
  - Have a fixed number of labels
  - Do not generalise well (at all)
  - CLIP overcomes these limitations

https://www.google.com/search?sca_esv=855ccbb775c3ea59&q=CLIP+original%3B+paper&tbm=vid&source=lnms&prmd=isvnbmtz&sa=X&ved=2ahUKEwi99tb33o6FAxXmT2wGHUeoBEQQ0pQJegQIHRAB&biw=1267&bih=680&dpr=2#vhid=2nKmbpIqP7_QiM&vssid=videos-aa511487

CLIP

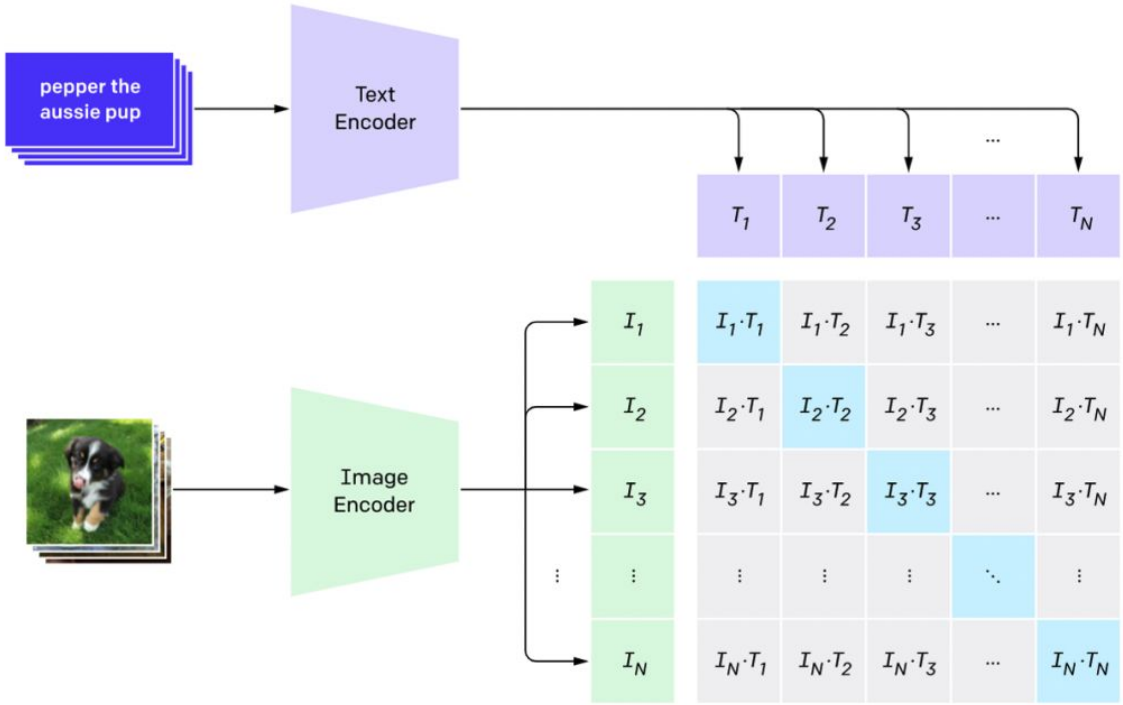
CLIP : Contrastive Language-Image Pre-training

- Model used for predicting which caption goes with which image
- Trained on 400 million Image-Text pairs collected from internet
- This pretraining with natural language supervision can be used for various downstream applications

Advantages of natural language supervision

- Much easier to scale as it doesn't require annotations as in "classic" ML compatible form
- Natural language is able to express, and therefore supervise, a much wider set of visual concepts
- Generality and usability increases so can be used for zero shot transfer
- ***Learning Transferable Visual Models From Natural Language Supervision by Radford et al - PMLR 2021***

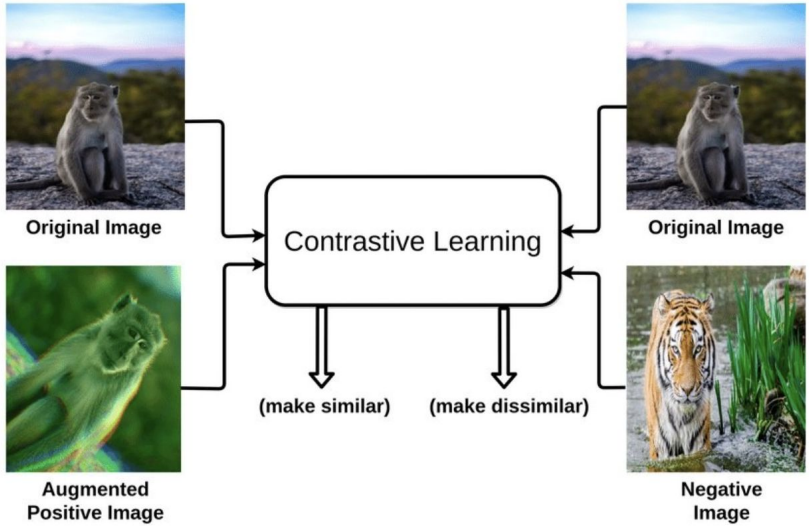
CLIP: Contrastive Language-Image Pretraining



- Model used for predicting which caption goes with which image
- Trained on 400 million Image-Text pairs collected from internet

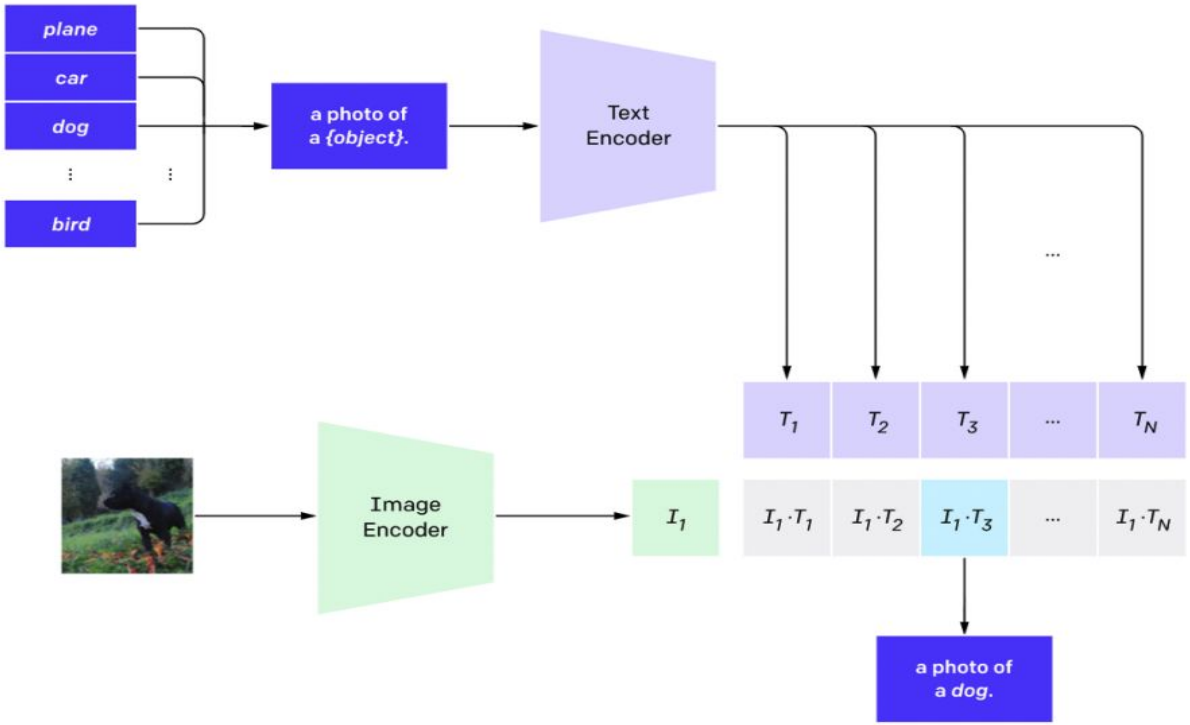
Credit: Prompting in Visual Intellignence

Contrastive Learning

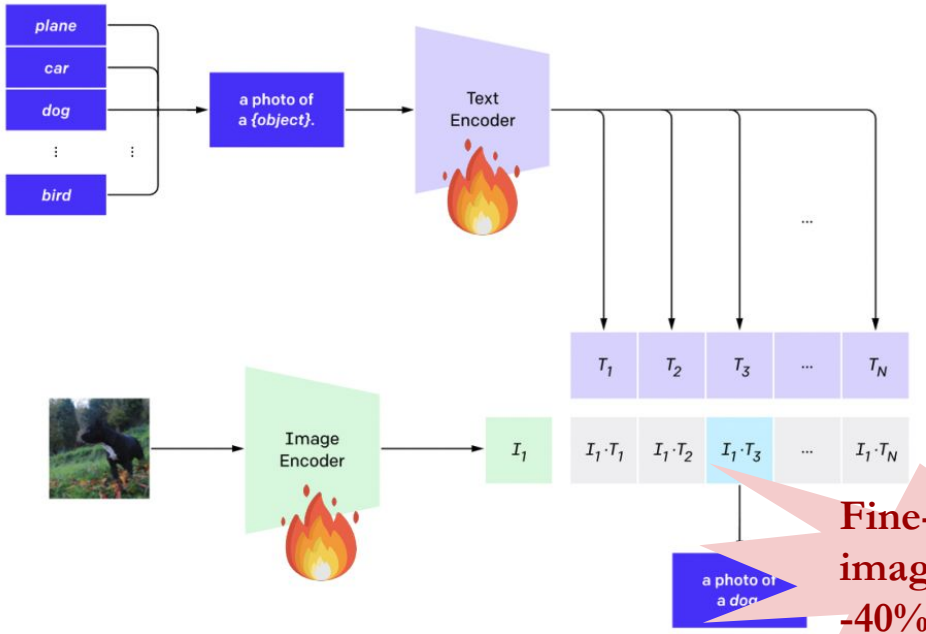


https://viso.ai/deep-learning/clip-machine-learning/

Zero shot Image Classification using CLIP



Fine-tuning?



Fine-tuning image encoder: -40%

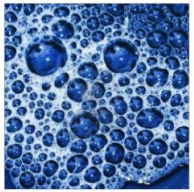
Fine-tune both: Collapse!

Credit: Prompting in Visual Intellignence

Prompt Engineering is hard though!

Caltech101	Prompt	Accuracy
	a [CLASS].	82.68
	a photo of [CLASS].	80.81
	a photo of a [CLASS].	86.29
	$[V]_1$ $[V]_2$ ... $[V]_M$ [CLASS].	**91.83**

Flowers102	Prompt	Accuracy
	a photo of a [CLASS].	60.86
	a flower photo of a [CLASS].	65.81
	a photo of a [CLASS], a type of flower.	66.14
	$[V]_1$ $[V]_2$ ... $[V]_M$ [CLASS].	**94.51**

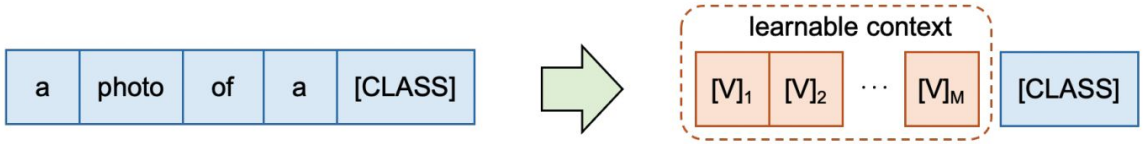
Describable Textures (DTD)	Prompt	Accuracy
	a photo of a [CLASS].	39.83
	a photo of a [CLASS] texture.	40.25
	[CLASS] texture.	42.32
	$[V]_1$ $[V]_2$ ... $[V]_M$ [CLASS].	**63.58**

EuroSAT	Prompt	Accuracy
	a photo of a [CLASS].	24.17
	a satellite photo of [CLASS].	37.46
	a centered satellite photo of [CLASS].	37.56
	$[V]_1$ $[V]_2$ ... $[V]_M$ [CLASS].	**83.53**

A slight change in wording could lead to big changes in performance!

Credit: Prompting in Visual Intellignence

Context Optimization (CoOp)



$$p(y=i|\boldsymbol{x}) = \frac{\exp(\cos(\boldsymbol{w_i}, \boldsymbol{f})/\tau)}{\sum_{j=1}^{K} \exp(\cos(\underline{\boldsymbol{w_j}}, \boldsymbol{f})/\tau)}$$

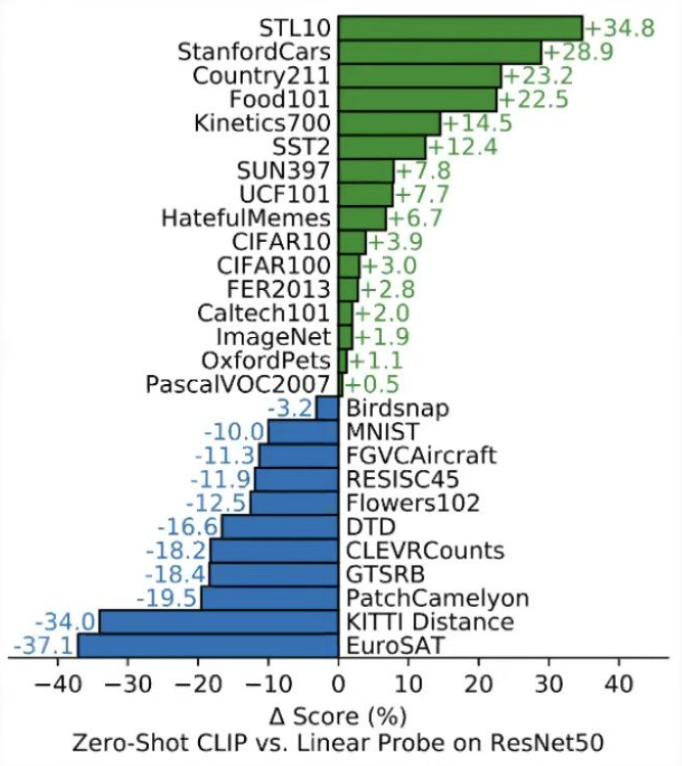
Hand-designed prompt

$$p(y=i|\boldsymbol{x}) = \frac{\exp(\cos(g(\boldsymbol{t}_i), \boldsymbol{f})/\tau)}{\sum_{j=1}^{K} \exp(\cos(g(\underline{\boldsymbol{t}_j}), \boldsymbol{f})/\tau)}$$

Learnable prompt

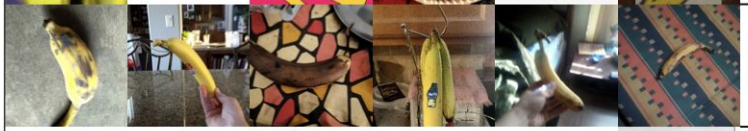
Zhou et al. "Learning to Prompt for Vision-Language Models", IJCV 2022

Zero-shot CLIP vs ResNet-50



- ResNet was pre-trained on ImageNet a
  linearly probed on each dataset
  - Basically just a tuned linear layer for each dat
- Zero-shot CLIP beats ResNet on 16/27
  - Including ImageNet
- New SoTA for STL10! (99.3%)
- General trend is that 'specialised' data
  perform worse with CLIP

Robustness to distribution shift

	Dataset Examples	ImageNet ResNet101	Zero-Shot CLIP	Δ Score
ImageNet		**76.2**	**76.2**	0%
ImageNetV2		64.3	**70.1**	+5.8%
ImageNet-R		37.7	**88.9**	+51.2%
ObjectNet		32.6	**72.3**	+39.7%
ImageNet Sketch		25.2	**60.2**	+35.0%
ImageNet-A		2.7	**77.1**	+74.4%

https://www.google.com/search?sca_esv=855ccbb775c3ea59&q=CLIP+original%3B+paper&tbm=vid&source=lnms&prmd=isvnbmtz&sa=X&ve